sion trees combine the strengths of two algorithms: regression trees (models that relate a response to their predictors by recursive binary splits) and boosting (an adaptive method for combining many simple models to give improved predictive performance). The final GBDT model can be understood as an additive regression model.[1]

Random forests are an ensemble learning method for classification and regression that operate by constructing a multitude of decision trees at training time and outputting the class that is the mode of the classes output by individual trees. The method combines "bagging" idea and the random selection of features in order to construct a collection of decision trees with controlled variance.[2]

The algorithm in details will be demonstrated in the third section - Proposed Method.

1.3. Evaluation

As we are aiming at continuous regression instead of discrete classification, MSE could be a better metric to evaluate rather than accuracy analysis. Furthermore, bias-variance decomposition will more specifically indicate the factor of whether the model is success or not. From we have discussed in class, a high bias can show the model's disability to learn relations between features effectively and hence would underfit on the dataset leading to low accuracy while predicting. By contrast, a high variance indicates the predicted output become highly flexible with respect to the data points of the dataset, which causes a model to overfit on the training data leading to low accuracy while predicting. So they are the key points considered for measuring success in our project, and we calculate according to the formulas as below.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

$$\begin{aligned} E[(y-\hat{y})^2] &= (y - E[\hat{y}])^2 + E[(E[\hat{y}] - \hat{y})^2] \\ &= Bias^2 + Variance \end{aligned}$$

2. Related Work

The subsections 10.10.2 and 10.10.3 of *The Elements of Statistical Learning* [3] mainly talked about the Gradient Boosting theory. It interprets the procedure of Gradient Boosting in Tree Algorithm, which uses base regression trees for each iteration, with residuals of the last iteration as input. It constructs a one-direction algorithm with optimization on every layer of iteration.

To realize the model, which R packages we mainly use are gbm[1] and randomForest[2]. They offer us the explicit applications and the meanings of all parameters of the function as well as the algorithm. Furthermore, they inspire us with how to tune hyperparameters.

The motivation of using GBDT is from the related works, so we conclude the good properties of the GBDT model which fits our dataset.

- It is based on the regression tree and applicable to our dataset which has a numeric **response variable**.
- It works for both categorical as well as numeric **feature values**.
- And it could prevent from overfitting.

Then we use the theory and information from R packages documentations to construct the models and make the comparison based on what we have learned and the citations [1] and [2].

3. Proposed Method

3.1. Data Processing

We have all kinds of data in the dataset where the most common form is JSON, which brought the initial problem.

(1) The lines in JSON are not in the same format with a variety of information that we cannot put them into a certain structure, such as 'browser' in Figure 1.

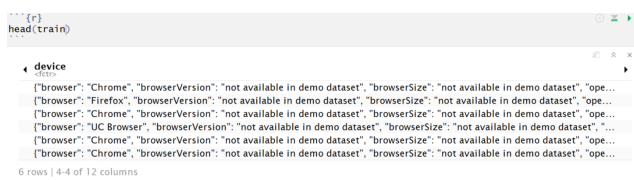

Figure 1. The glance of the training set

We use the function fromJSON of the package "jsonlite" to dissolve the layers of braces "{}" and construct a clear structure. The codes of extract process are shown in Figure 2 and Figure 3.

[1] https://cran.r-project.org/web/packages/gbm/gbm.pdf
[2] https://cran.r-project.org/web/packages/randomForest/randomForest.pdf